

Rolling-burden alarm and temporal granularity

1. What the alarm is

The alarm is a **prospective rolling-burden** rule on behavioural topic probabilities from LDA / DCABP.

At each decision time (t):

1. Build **pUF** = $P(\text{topic 3}) + P(\text{topic 4}) + P(\text{topic 5})$ (unfavourable day-types).
2. Aggregate pUF over a **lookback window W** into a **burden** score.
3. Raise an alarm if $\text{burden} \geq \text{threshold } (\theta)$.
4. Label the decision as positive if progression (PD) occurs within a future **horizon H** after (t).

Only decisions with enough history (full W) and enough follow-up to score the label (or a known PD time) are kept. Those eligible ((patient, t)) pairs are the **decision points**.

2. Parameters that can be varied

Parameter	Role	Typical choices
W (lookback)	How much past burden is summarised at (t)	60 / 90 / 120 days ($\approx 2 / 3 / 4$ months)
H (horizon)	How far ahead PD counts as a hit	90 / 120 / 180 days ($\approx 3 / 4 / 6$ months)
Granularity	How often decisions are made and how LDA is built	daily, weekly, biweekly, monthly
Score family	How pUF / burden are constructed	sampled vs collapsed (below)
θ (threshold)	Alarm cutoff on the burden scale	Youden-optimal, or a fixed operating point (e.g. ~ 1.61 on monthly <i>sum</i> scale)
Event / censor time	Clock used to stop landmarks and label PD	clinical follow-up days (<code>Obs_time</code>) or eB2-aligned event time

Sampled vs collapsed

Family	Idea
Sampled	Daily sliding LDA (30-day window) → daily pUF; weekly / biweekly / monthly only change <i>when</i> you decide (same underlying daily signal). Burden = rolling mean over the last W days.
Collapsed	One LDA per non-overlapping block (7 / 14 / 30 d) → block pUF; burden = rolling mean over the last <code>round(W/block)</code> blocks.

Collapsed block pUF is **not** the mean of daily pUF inside the block: each family is its own LDA construction.

Monthly collapsed landmarks

On the monthly scale, month (m) is placed at day ($t = m \times 30$). A landmark is kept only if (t) is strictly before the event/censor time. Burden is the **sum** of pUF over the last W months (so thresholds are on a sum scale, ~1.61, not the mean scale ~0.5 used elsewhere).

3. Experiment in this report

We fix a main operating region and then sweep the knobs above:

1. **Monthly collapsed baseline** — $W = 3$ months, $H = 4$ months, clinical `obs_time` as event/censor clock, Youden / fixed $\theta \approx 1.61$.
2. **Granularity comparison** — $W = 90$ d, $H = 120$ d; sampled (daily→monthly) vs collapsed (weekly / biweekly / monthly).
3. **$W \times H$ grid** — $W \in \{60, 90, 120\}$ d $\times H \in \{90, 120, 180\}$ d.
4. **Individual trajectories** — example patients with and without PD.

Outputs live under `results/alarm/` (`tables/`, `figures/`, `analysis_figures/`, `analysis_outputs/`). Primary monthly landmarks: `tables/decision_points_W3_H4.csv` (**Obs_time**, $n=1015$).

Delivery summary figures (high-res, Obs_time-aligned, AUROC with patient-bootstrap CIs): [analysis_figures/](#).

4. Results — monthly collapsed baseline ($W = 3$ mo, $H = 4$ mo)

Score = **sum** of pUF over W months; cutoff: `month  $\times$  30 < obs_time`.

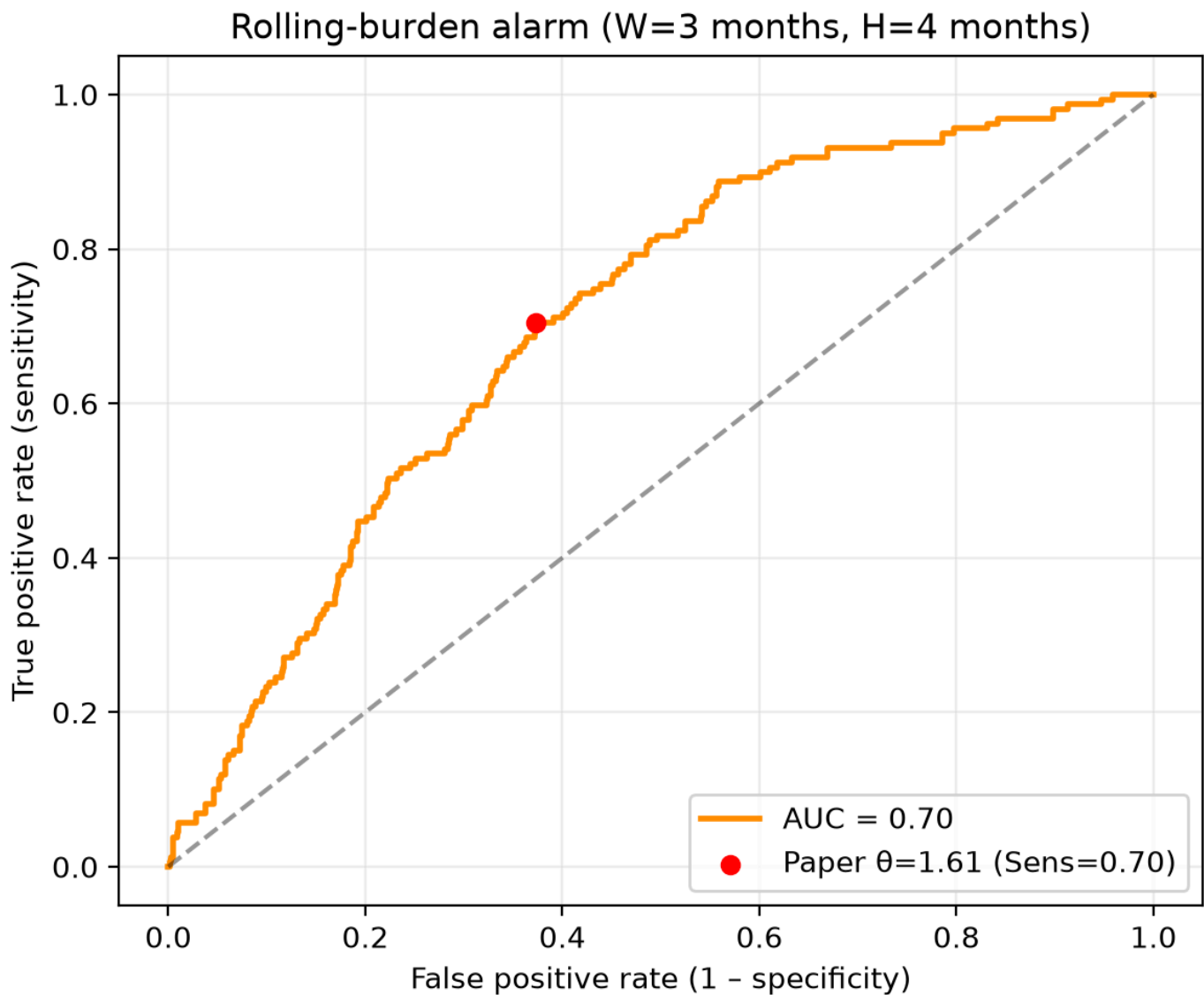
Metric	Value
ROC-AUC	0.704

Metric	Value
Sens / Spec ($\theta \approx 1.61$, sum scale)	0.704 / 0.626
PPV / NPV	0.259 / 0.919
TP / FN / FP / TN	112 / 47 / 320 / 536

Threshold scale (same rule, two writings):

Scale	θ	Relation
Sum (as used here)	1.61	sum of last W=3 monthly pUF
Mean (equivalent)	≈ 0.537	1.61 / 3

Youden exact: 1.610762 (sum) $\Leftrightarrow$ **0.536921** (mean). AUC is identical under either writing; only the threshold number changes. The mean form (~ 0.54) is on the same ~ 0 – 1 scale as sampled / other collapsed configs.



5. Results — granularity (W = 90 d, H = 120 d)

Operating point = Youden threshold per row.

Sampled and non-monthly collapsed use a **mean** burden (~0.4–0.6). Monthly collapsed is reported on the **sum** scale (~1.61); mean-equivalent $\theta \approx 0.537$ (see §4).

Clinical clock: all rows below use the same follow-up field as the monthly baseline — **obs_time** (sampled/collapsed pUF aligned to `Subjects_data.obs_time` before scoring).

AUROC uncertainty: 95% CIs from **patient-clustered bootstrap** (resample patients with replacement; 2000 reps). See `analysis_figures/auc_ci_W90_H120.csv`.

Granularity	n	AUC	95% CI	Sens	Spec	PPV	NPV
daily (sampled)	36316	0.684	0.611–0.751	0.643	0.646	0.240	0.912

Granularity	n	AUC	95% CI	Sens	Spec	PPV	NPV
weekly sampled	5235	0.683	0.610–0.750	0.645	0.646	0.239	0.913
biweekly sampled	2651	0.680	0.607–0.746	0.674	0.613	0.230	0.917
monthly sampled	1278	0.682	0.610–0.747	0.827	0.475	0.210	0.942
weekly collapsed	4127	0.707	0.625–0.780	0.753	0.590	0.251	0.929
biweekly collapsed	2125	0.695	0.614–0.767	0.713	0.601	0.259	0.914
monthly collapsed (30 d)	1015	0.704	0.632–0.777	0.704	0.626	0.259	0.919

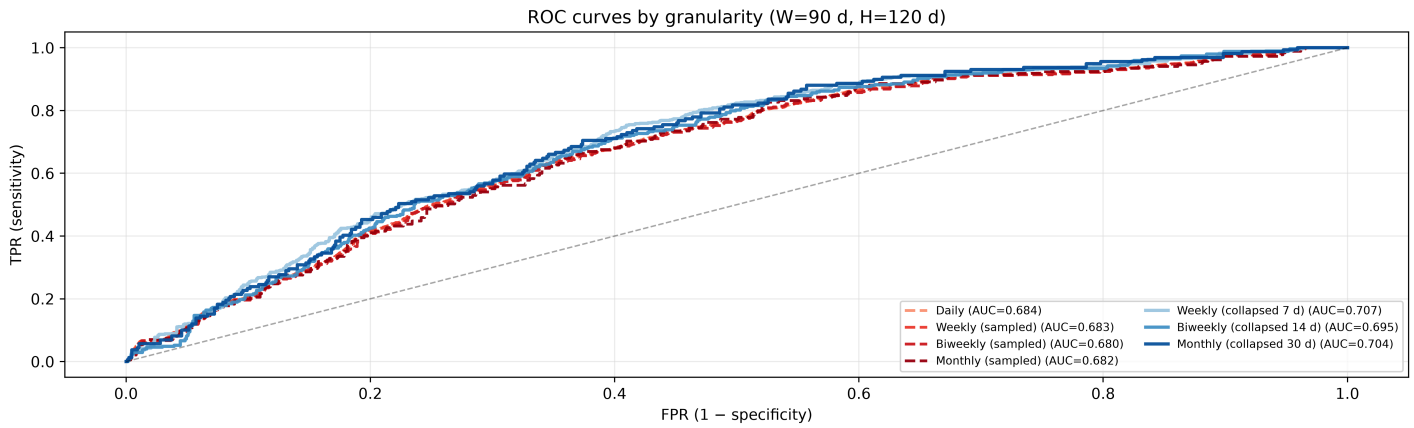
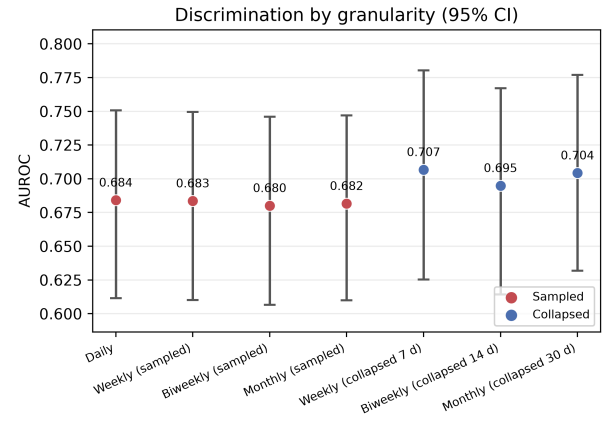
Short read

- **Sampled** (daily→monthly): AUC nearly flat (~0.68). Subsampling does not improve discrimination; it only reduces n.
- **Collapsed** improves somewhat vs sampled (~0.70–0.71). Highest point estimate here: **weekly collapsed (AUC 0.707)**; CIs overlap across collapsed configs.
- PPV remains low (~0.21–0.26); NPV high (~0.91–0.94).
- Do not compare monthly $\theta \approx 1.61$ to sampled $\theta \approx 0.55$ without converting: use $\theta_{\text{mean}} = \theta_{\text{sum}} / W$.

High-resolution delivery panel (metrics table + AUROC points with 95% CI + ROC; no winner star):

Alarm performance across temporal granularities
Metrics at Youden threshold (W=90 d, H=120 d)

Granularity	N	AUROC	Sens	Spec	PPV	NPV
Daily	36316	0.684	0.643	0.646	0.240	0.912
Weekly · sampled	5235	0.683	0.645	0.646	0.239	0.913
Biweekly · sampled	2651	0.680	0.674	0.613	0.230	0.917
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6. Results — W × H grid

Grid: $W \in \{60, 90, 120\} \text{ d} \times H \in \{90, 120, 180\} \text{ d}$.

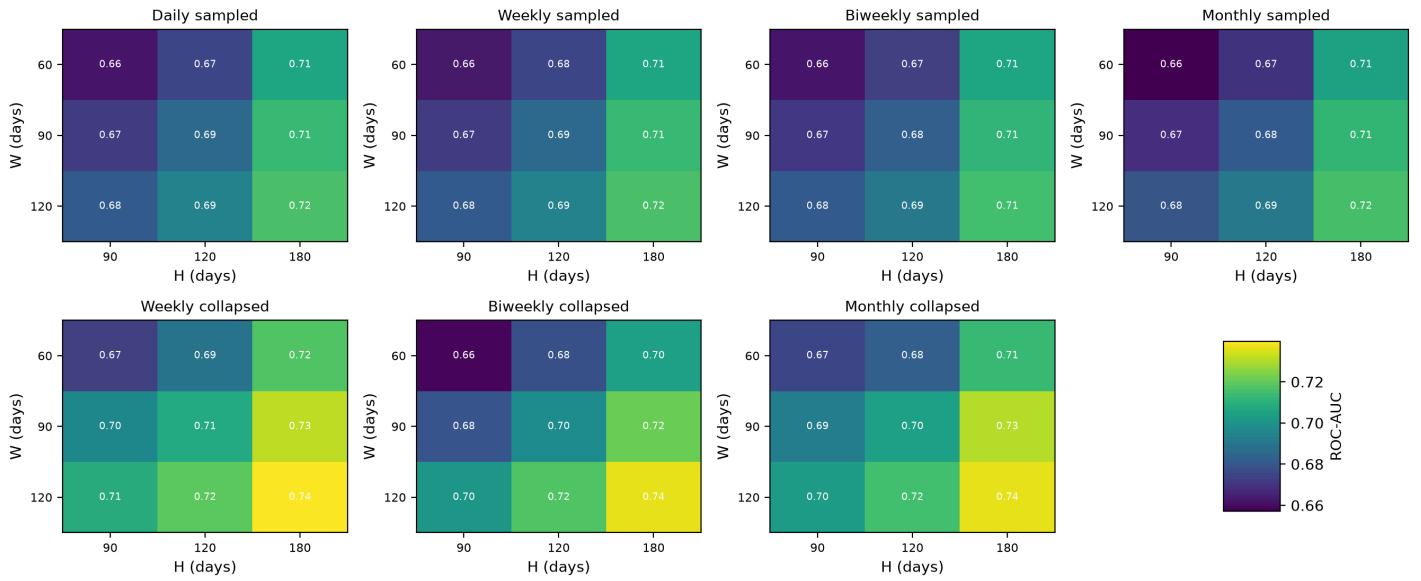
AUC (selected)

	H=90	H=120	H=180
Daily W=60/90/120	0.662 / 0.672 / 0.681	0.675 / 0.686 / 0.694	0.706 / 0.713 / 0.716
Weekly collapsed ★	0.673 / 0.697 / 0.709	0.688 / 0.707 / 0.719	0.717 / 0.732 / 0.740
Monthly collapsed	0.675 / 0.692 / 0.702	0.683 / 0.704 / 0.715	0.713 / 0.730 / 0.737

Read

- $\uparrow H \rightarrow$ AUC rises monotonically.
- $\uparrow W \rightarrow$ AUC rises moderately (more historical load).
- Ranking **collapsed** $\geq$ **sampled** holds across almost the whole grid.
- $W \approx 90$ / $H \approx 120$ sits in the mid region of the map.

Mini experiment: ROC-AUC across lookback W × horizon H



AUC vs prediction horizon H (curves = W lookback)

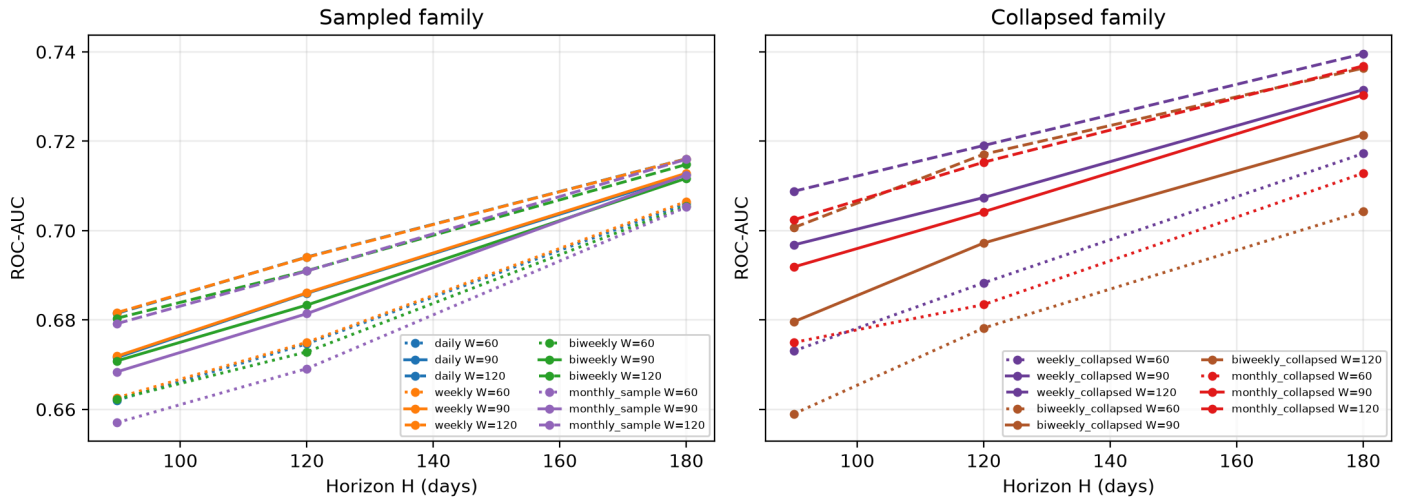


Table: [tables/wh_grid_auc.csv](#)

7. Results — individual trajectories

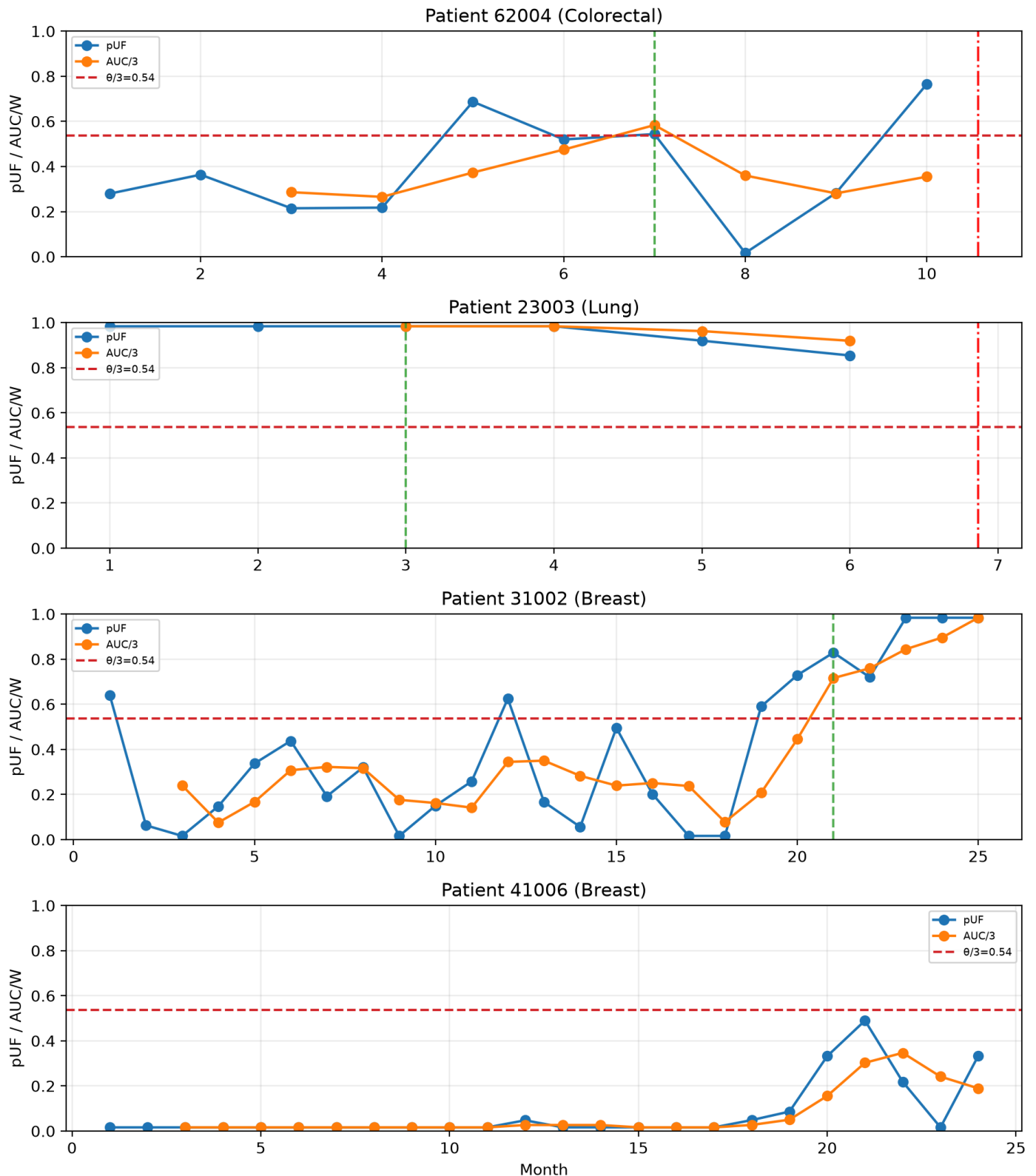
Patients: 62004, 23003, 31002, 41006.

ID	Profile	Notes
62004	PD (day 317)	1st alarm ~month 7 (lead before PD)
23003	PD (day 206)	High burden from the start; early alarm
31002	No PD	Late alarm (~month 21) with no event in follow-up
41006	No PD	Below threshold; no alarm

7.1 Monthly collapsed (pUF + rolling burden)

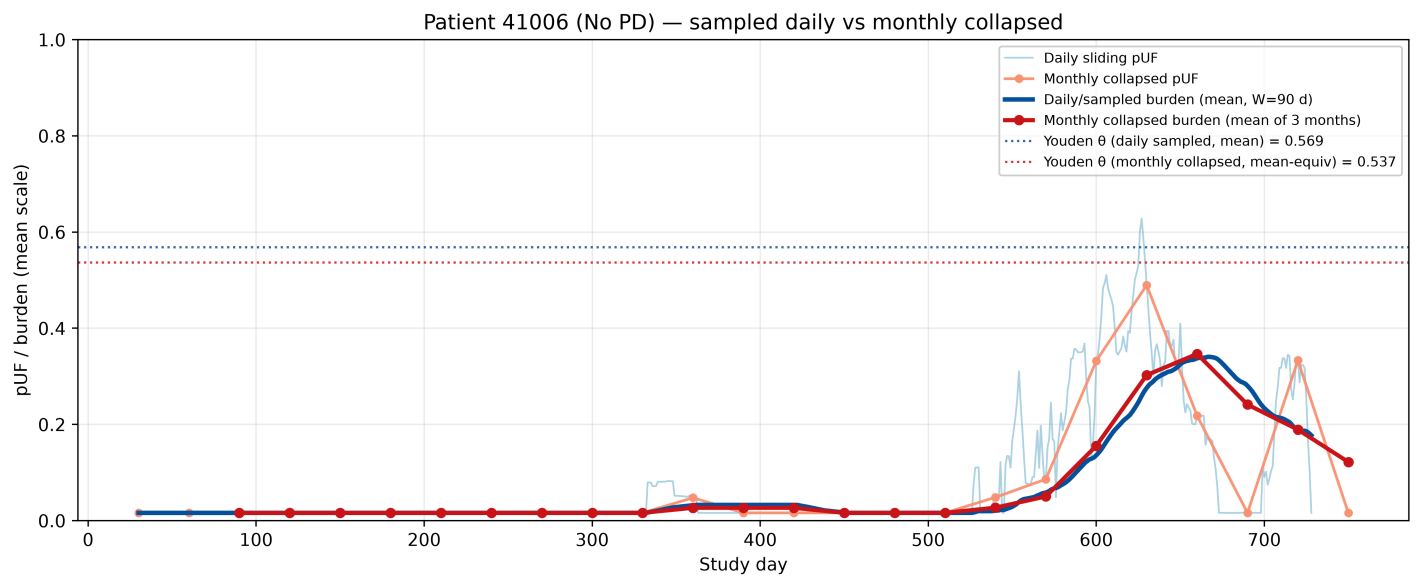
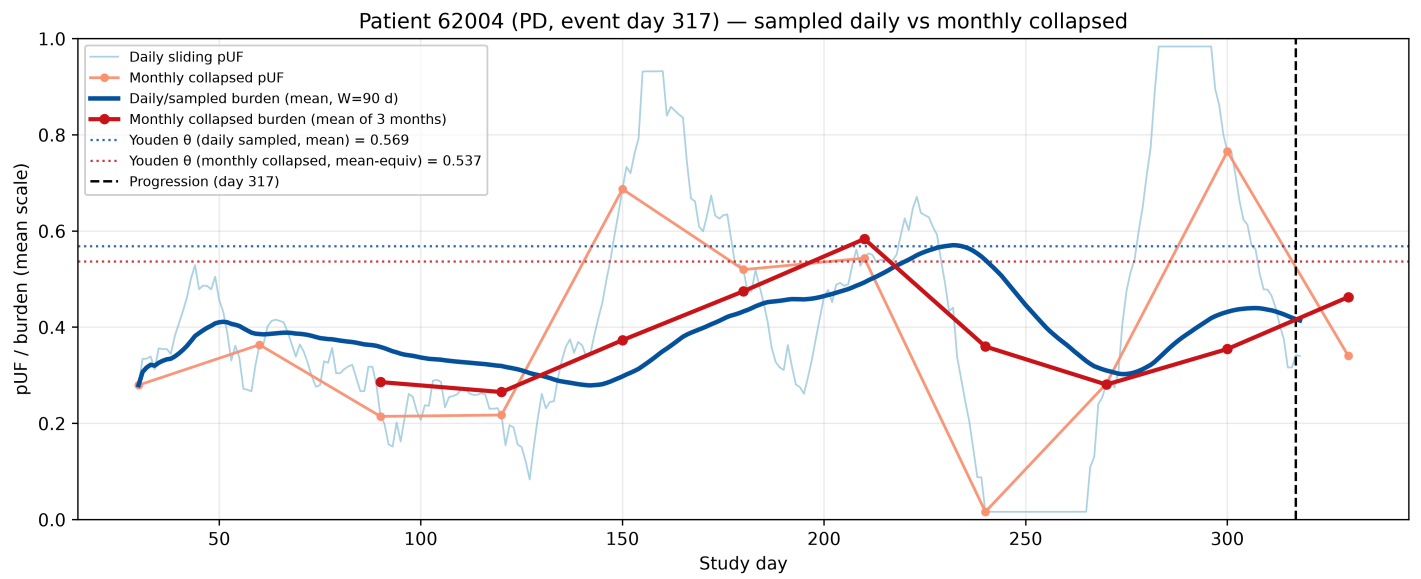
W=3 months, H=4 months, $\theta=1.61$ (mean-scale $\theta/W \approx 0.54$).

Paper-style trajectories — monthly pUF & rolling AUC (W=3, H=4, $\theta=1.61$)



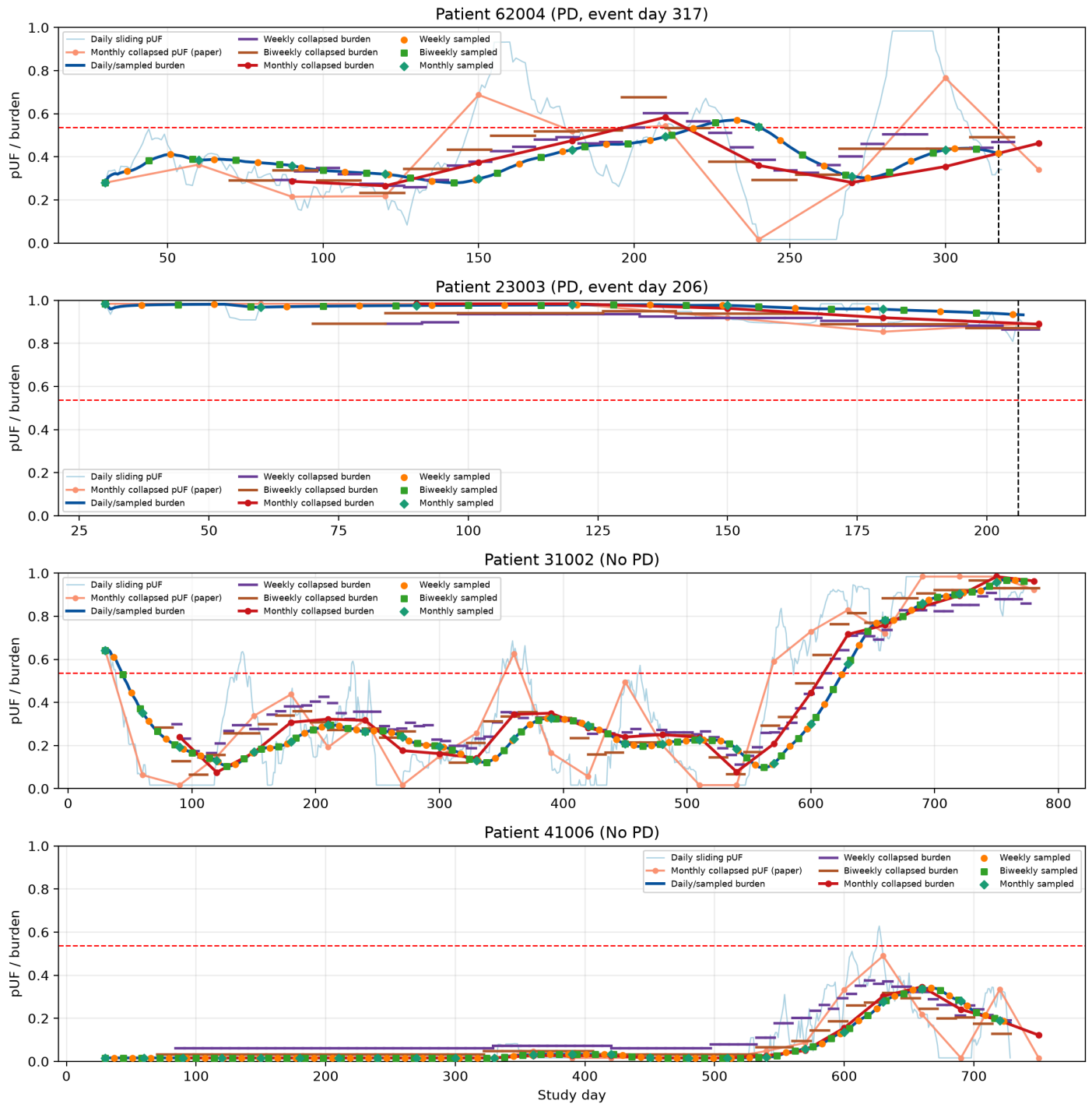
7.2 Sampled vs collapsed (W = 90 d)

Simplified high-res delivery plots (daily/sampled burden vs monthly collapsed; Obs_time clock; construction-specific Youden θ labelled):



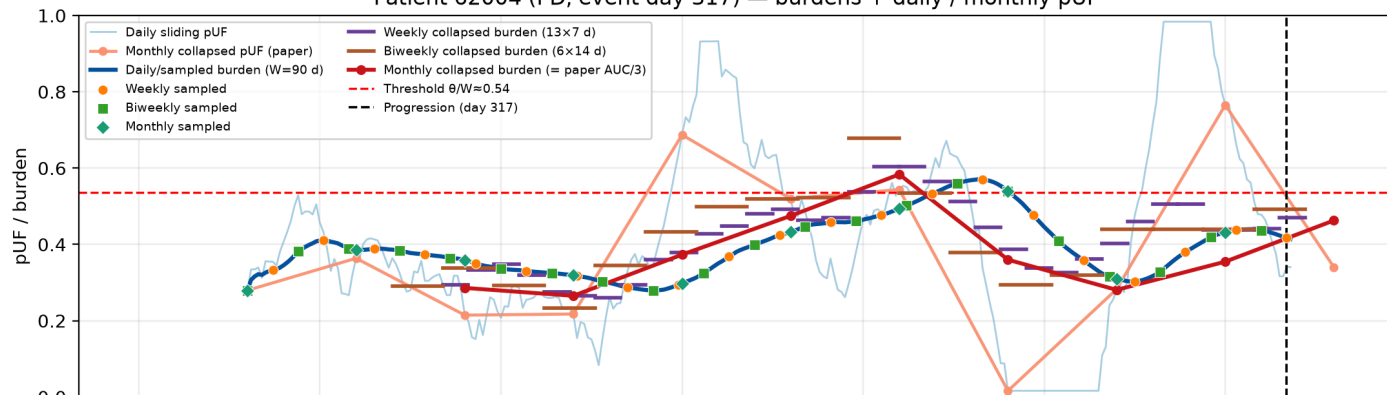
Full multi-panel trajectories:

Burdens by construction + daily / monthly-collapsed pUF (sampled share daily burden; collapsed recomputes its own)

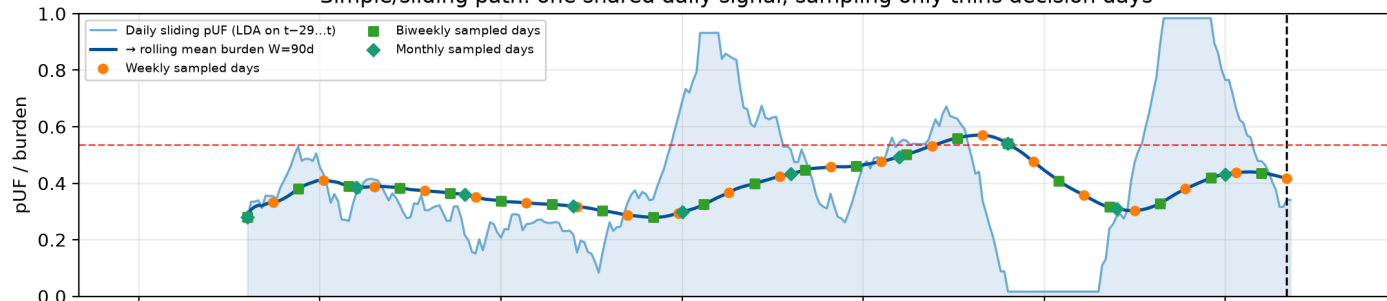


- **Blues:** daily sliding pUF + daily/sampled burden
- **Reds:** monthly collapsed pUF + monthly collapsed burden
- Steps: weekly / biweekly collapsed burden
- Markers: sampled evaluation (7 / 14 / 30 d) on the daily burden

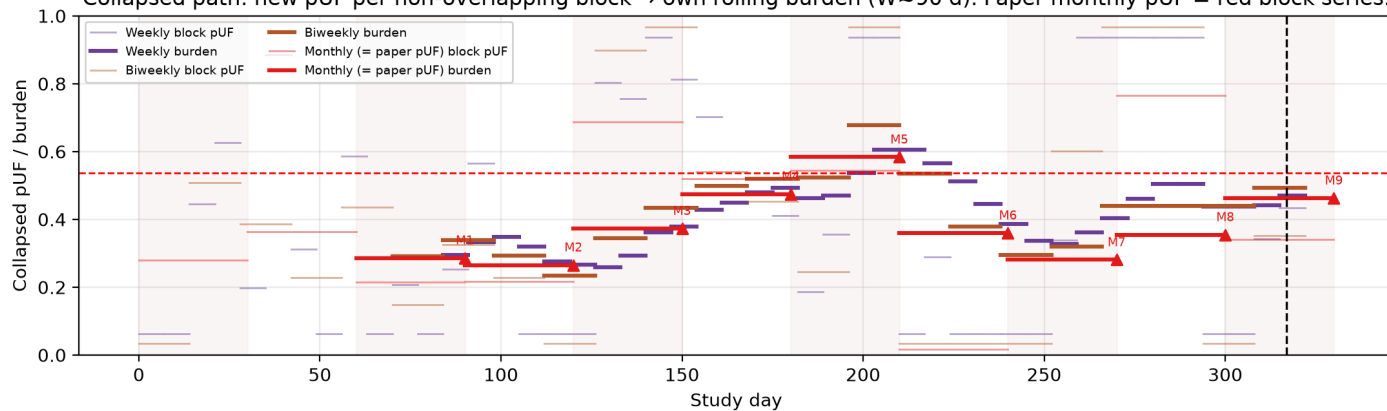
Patient 62004 (PD, event day 317) — burdens + daily / monthly pUF



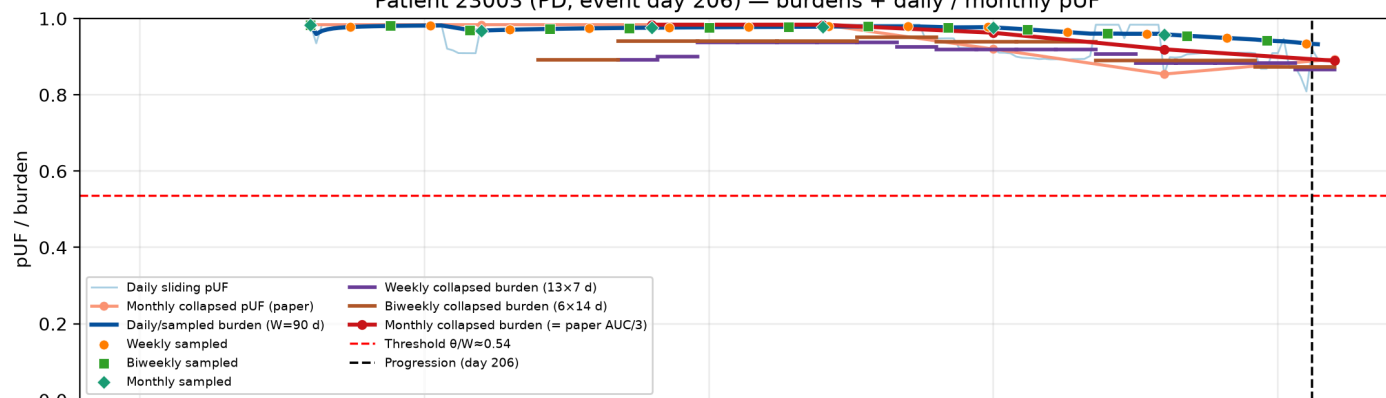
Simple/sliding path: one shared daily signal; sampling only thins decision days



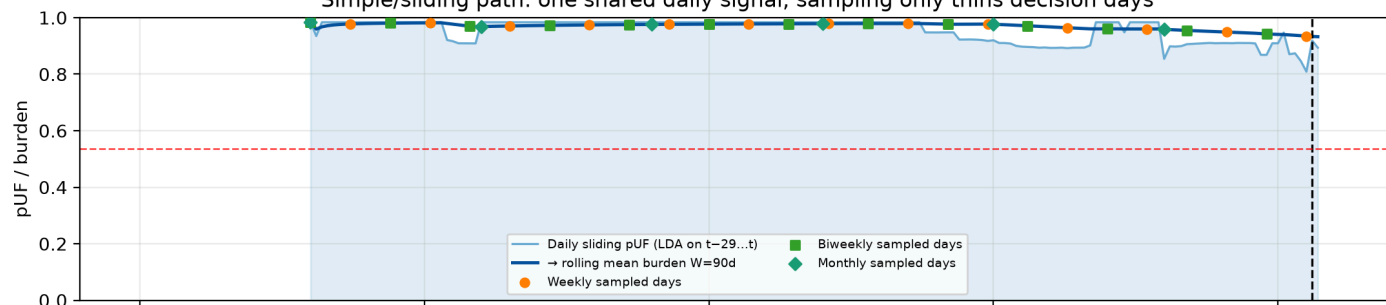
Collapsed path: new pUF per non-overlapping block $\rightarrow$ own rolling burden ($W \approx 90$ d). Paper monthly pUF = red block series.



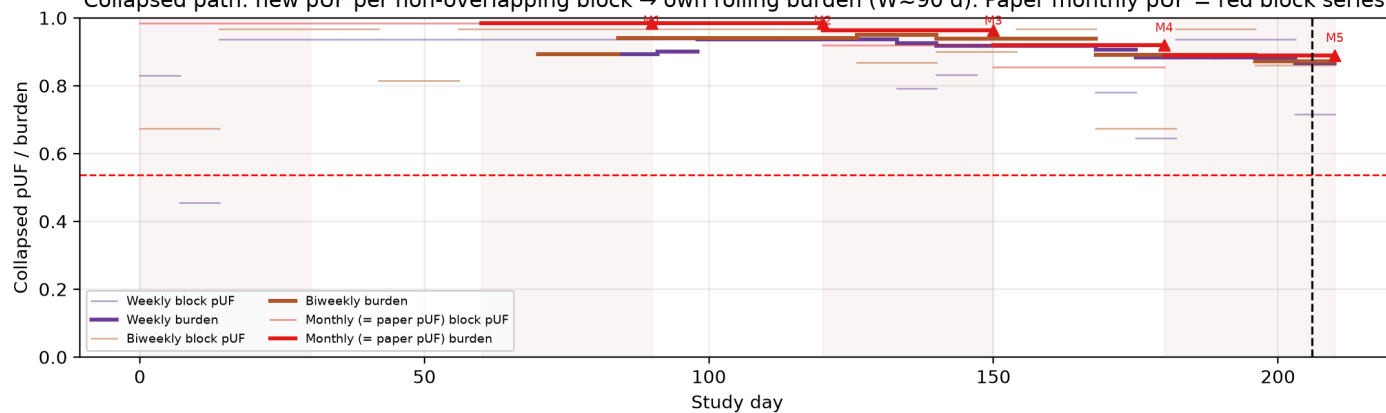
Patient 23003 (PD, event day 206) — burdens + daily / monthly pUF



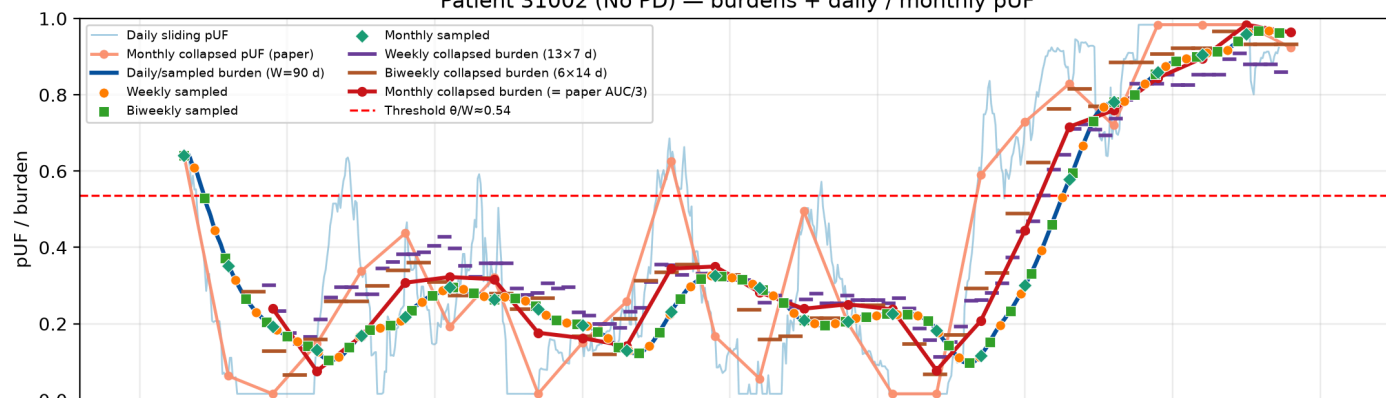
Simple/sliding path: one shared daily signal; sampling only thins decision days



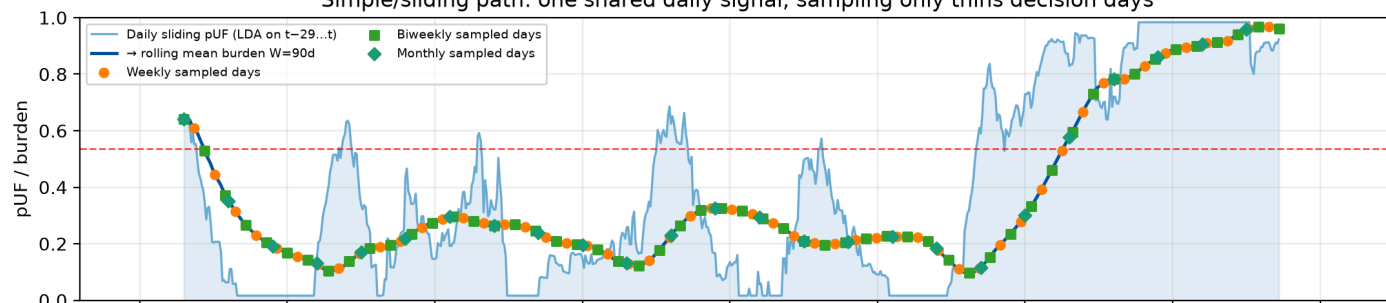
Collapsed path: new pUF per non-overlapping block → own rolling burden (W≈90 d). Paper monthly pUF = red block series.



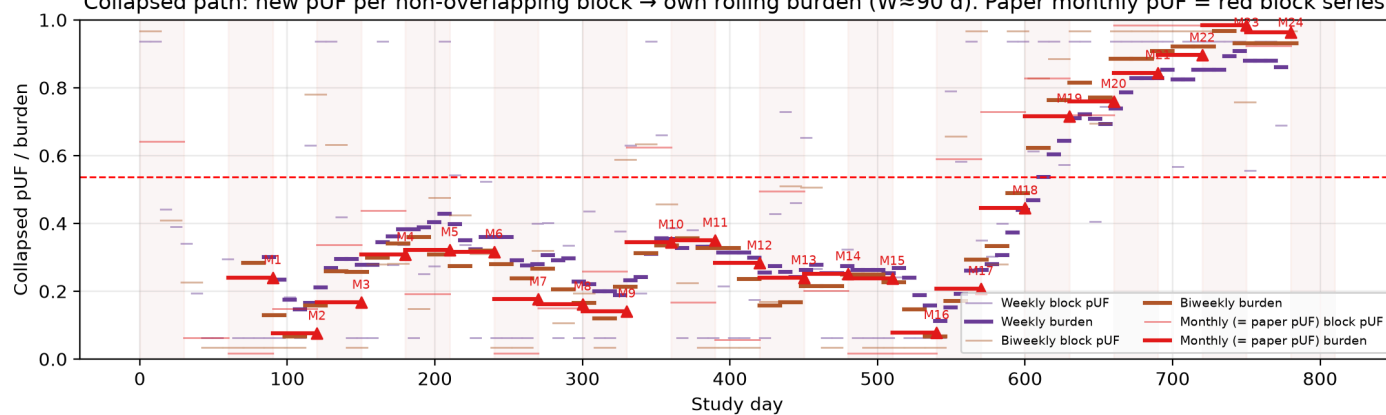
Patient 31002 (No PD) — burdens + daily / monthly pUF

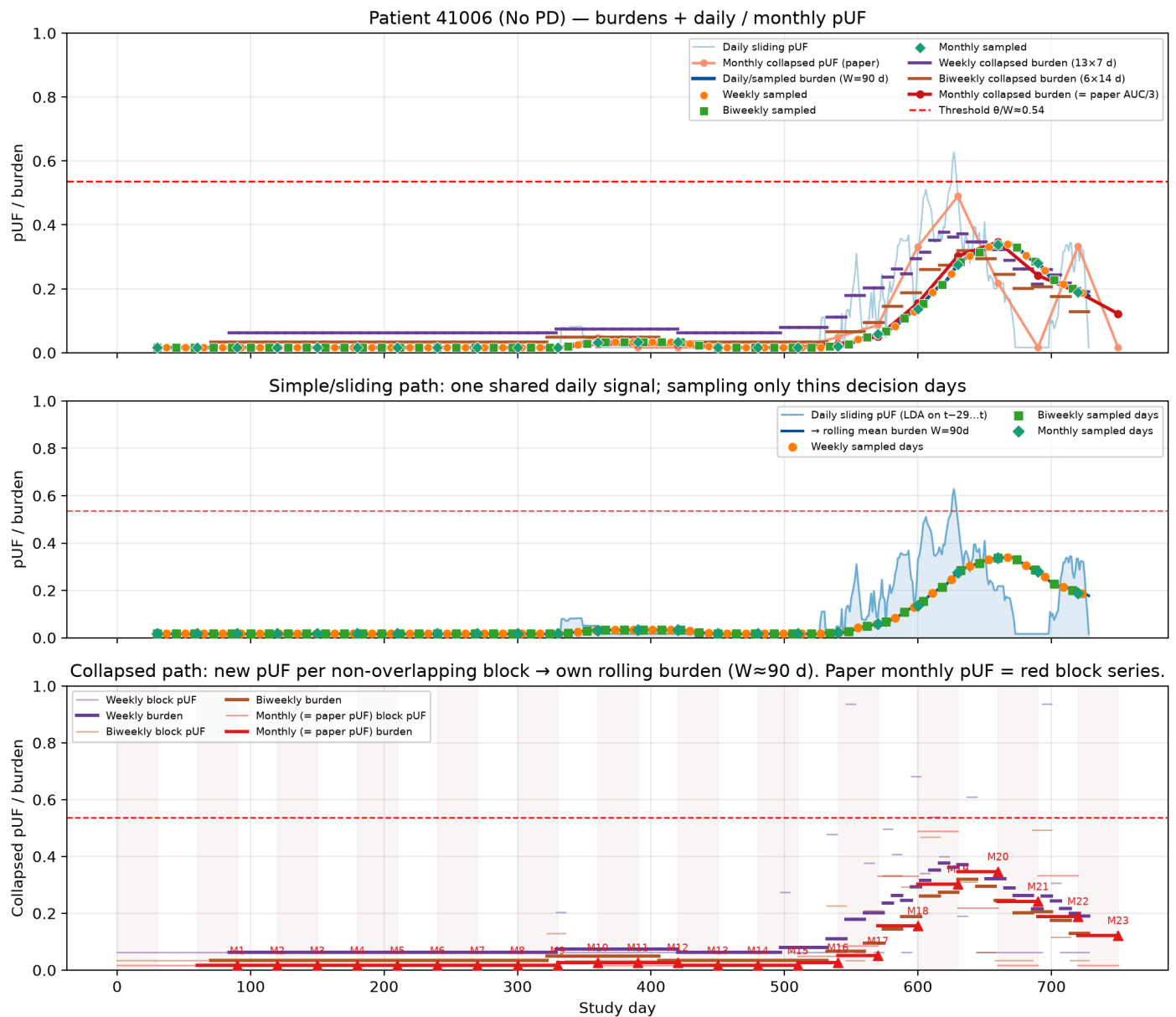


Simple/sliding path: one shared daily signal; sampling only thins decision days



Collapsed path: new pUF per non-overlapping block → own rolling burden (W≈90 d). Paper monthly pUF = red block series.





8. How to reproduce

Monthly baseline + ROC sweep

```
.venv/bin/python scripts/03_analysis/alarm/run_granularity_sweep.py --monthly-only
```

```
.venv/bin/python scripts/03_analysis/alarm/run_granularity_sweep.py --skip-lda
```

WxH mini-experiment (+ heatmaps)

```
.venv/bin/python scripts/03_analysis/alarm/analyze_missingness_and_wh_grid.py
```

Monthly trajectories

```
.venv/bin/python scripts/03_analysis/alarm/plot_patient_trajectories.py \
```

```
--patients 62004,23003,31002,41006
```

Sampled vs collapsed trajectories

```
.venv/bin/python scripts/03_analysis/alarm/plot_granularity_trajectories.py \
```

```
--patients 62004,23003,31002,41006 --W 90
```

```
# High-res delivery figures (Obs_time-aligned; AUROC patient-bootstrap CIs)  
.venv/bin/python scripts/03_analysis/alarm/plot_analysis_figures.py
```

File map: [README.md](#) . Delivery panel: [analysis_figures/](#) .

9. Conclusions

1. **Subsampling** the same daily signal (sampled) **does not improve** discrimination: AUC stays **~0.68** from daily→monthly.
2. **Collapsing** LDA into non-overlapping blocks (7/14/30 d) **does improve** somewhat: AUC **~0.70–0.71** at W=90 d, H=120 d.
3. The **highest point estimate** in that setting is **weekly collapsed** (AUC **0.707**; Youden Sens/Spec $\approx 0.75/0.59$); patient-bootstrap 95% CIs overlap across collapsed configs.
4. **↑W** and **↑H** raise AUC in an ordered way; ranking collapsed $\geq$ sampled holds on the grid.
5. **PPV** stays low (**~0.25**): the alarm discriminates but yields many false positives at useful thresholds.
6. Individually, the score can anticipate PD (lead time), mark persistently high burden, or fire late alarms with no event — useful for interpretation beyond aggregate metrics.
7. The delivery summary figure uses the **same clinical follow-up field (obs_time)** for all granularities as the monthly baseline (n=1015), with **AUROC 95% CIs from patient-level bootstrap**.

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